QQ-A-250/18F
September 21, 1978
SUPERSEDING
QQ-A-250/18E
February 24, 1971

FEDERAL SPECIFICATION

# ALUMINUM ALLOY ALCLAD ONE SIDE 7075, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy alclad one side 7075 plate and sheet; the general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7075) core with a thin layer of an aluminum alloy (7072) anodic to the core bonded to one surface.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as 0, T6, T62, T651 or F temper, as specified (see 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

Federal Standard:

Fed. Std. No. 184 - Identification Marking of Aluminum, Magnesium, and Titanium.

FSC 9535

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Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Core and cladding. The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the alclad plates and sheets shall conform to the requirements specified in table I for core and cladding, respectively.

TABLE I. Chemical composition 1/

| Element | Analysis | | | |
|---|---|---|---|---|
| | Core(7075) | | Cladding (7072) | |
| | Percent | | Percent | |
| | Minimum | Maximum | Minimum | Maximum |
| Zinc | 5.1 | 6.1 | 0.8 | 1.3 |
| Magnesium | 2.1 | 2.9 | - | 0.10 |



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TABLE I. Chemical composition 1/

| Element | Analysis | | | |
|---|---|---|---|---|
| | Core(7075) | | Cladding (7072) | |
| | Percent | | Percent | |
| | Minimum | Maximum | Minimum | Maximum |
| Copper | 1.2 | 2.0 | - | 0.10 |
| Chromium | 0.18 | 0.28 | - | - |
| Manganese | - | 0.30 | - | 0.10 |
| Iron | - | 0.50 | - | 2/ |
| Silicon | - | 0.40 | - | 2/ |
| Titanium | - | 0.20 | - | - |
| Others, each | - | 0.05 | - | 0.05 |
| Others, total | - | 0.15 | - | 0.15 |
| Aluminum | Remainder | | Remainder | |

1/ Analysis shall be made regularly only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

2/ Iron plus silicon: 0.7 percent maximum.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties of the material perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to table II for the temper specified.



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TABLE II. Mechanical properties

| Temper | Thickness | Tensile strength, minimum | Yield strength at 0.2 percent offset, minimum | Elongation in 2 in. or 4 times D 1/, 2/, minimum |
|---|---|---|---|---|
| | Inches | psi | psi | Percent |
| 0 | 0.015 thru 0.062 | 3/38,000 | 3/21,000 | 10 |
| | 0.063 thru 0.187 | 3/39,000 | 3/21,000 | 10 |
| | 0.188 thru 0.499 | 3/39,000 | 3/21,000 | 10 |
| | 0.500 thru 1.000 | 4/,3/40,000 | -- | 10 |
| T6 and T62 5/ | 0.012 thru 0.039 | 73,000 | 63,000 | 7 |
| | 0.040 thru 0.062 | 74,000 | 64,000 | 8 |
| | 0.063 thru 0.187 | 75,000 | 65,000 | 8 |
| | 0.188 thru 0.249 | 76,000 | 66,000 | 8 |
| T651 and T62 5/ | 0.250 thru 0.499 | 76,000 | 66,000 | 9 |
| | 0.500 thru 1.000 | 4/78,000 | 4/68,000 | 7 |
| | 1.001 thru 2.000 | 4/77,000 | 4/67,000 | 6 |
| F | All | 6/ | 6/ | 6/ |

1/ Not required for material 1/2 inch or less in width.

2/ D represents specimen diameter.

3/ Maximum.

4/ These properties are those of the core material inasmuch as the tests are made on a round specimen machined from the plate. The minimum tensile and yield strength across the whole section of the plate will be about 1 1/2 percent lower than these values.

5/ Material in the T62 temper is not available from the materials producers.

6/ No requirements.

3.2.2 Mechanical properties after heat-treatment. In addition to conforming to 3.2.1, material in the annealed (0) and the as-fabricated (F) tempers, without the subsequent imposition of cold work or forming operations, shall after proper solution heat-treatment and aging, also conform to the requirements of table II for the T62 temper. Material as received in the T6 and T651 tempers shall, after proper re-solution heat-treatment and aging, be capable of conforming to the requirements specified in table II for the T62 temper.



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3.2.3 Bend test. Bend test specimens taken from material, shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

TABLE III. Bend test factor "N"

| Thickness | Tempers | |
|---|---|---|
| | 0 | T6 and T62 |
| Inch | | |
| 0.012 thru 0.020 | 1 | 7 |
| 0.021 thru 0.032 | 1 | 7 |
| 0.033 thru 0.063 | 2 | 8 |
| 0.064 thru 0.091 | 3 | 9 |
| 0.092 thru 0.125 | 4 | 10 |
| 0.126 thru 0.249 | 5 | 11 |
| 0.250 thru 0.499 | 6 | 13 |

3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7072) plates that are bonded to one side of the aluminum alloy (7075) ingot or slab to form a composite that is to be rolled to the finished thickness, shall have a thickness as specified in table IV.

TABLE IV. Cladding thickness

| Thickness of finished plate or sheet | Average thickness of cladding on the finished plate or sheet; minimum percent of plate or sheet thickness |
|---|---|
| Inches | |
| Under 0.063 | 3.2 |
| 0.063 thru 0.187 | 2 |
| 0.188 and over | 1/ 1.2 |

1/ For plate 0.500 inch and over, the average cladding thickness shall have a maximum value of 3 percent of the plate thickness.

3.3.2 Thickness of cladding. If question arises concerning the thickness of cladding of the finished sheet or plate, samples examined in accordance with QQ-A-250 shall show an average thickness of cladding not less than that specified in table IV.



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3.4 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

TABLE V. Ultrasonic discontinuity acceptance limits 1/

| Thickness | Maximum weight per piece | Discontinuity class 2/ |
|---|---|---|
| Inches | Pounds | |
| 0.500 thru 1.499 | 2,000 | B |
| 1.500 thru 2.000 | 2,000 | A |

1/ Discontinuities in excess of those listed in table V may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas.

2/ The discontinuity class limits are defined in MIL-I-8950.

3.5 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250.)

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests after heat treatment. From material in the annealed (O) and asfabricated (F) tempers, an additional number of specimens equal to that required by QQ-A250 shall be taken and tested after solution heat-treatment and artificial aging to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY (See QQ-A-250.)

6. NOTES

6.1 Intended use. Alclad one side 7075 plates and sheets are used where very high strength with good resistance to corrosion is required, and where fabrication into aircraft skins or similar structures involves chemical milling of the core alloy. Formability in the heat treated and aged condition is not as good as for lower strength aluminum alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:



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(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).
(c) Dimensions required.
(d) Requirements for sizes not specifically covered (see QQ-A-250).
(e) Whether ultrasonic inspection of plate is required (see 3.4).
(f) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Temper. Temper T651 is available in plate only.

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